

XIAMEN UNIVERSITY MALAYSIA
厦門大學馬來西亞分校

COM105 Engineering Physics (I)

Lecture 10: Temperature and Heat

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In this chapter, you'll learn…

- the meaning of thermal equilibrium, and what thermometers really measure.
- the physics behind the absolute, or Kelvin, temperature scale.
- how the dimensions of an object change as a result of a temperature change.
- how to do calculations that involve heat flow, temperature changes, and changes of phase.
- how heat is transferred by conduction, convection, and radiation.

Thermodynamics 热力学

- The study of energy transformations involving heat, mechanical work, and other aspects of energy and how these transformations relate to the properties of matter. 物质
- Thermodynamics forms an indispensable part of the foundation of physics, chemistry, and the life sciences, and its applications turn up in such places as car engines, refrigerators, biochemical processes, and the structure of stars. 必不可少
- The terms “temperature” and “heat” have very different meanings, even though most people use them interchangeably. 可交换的

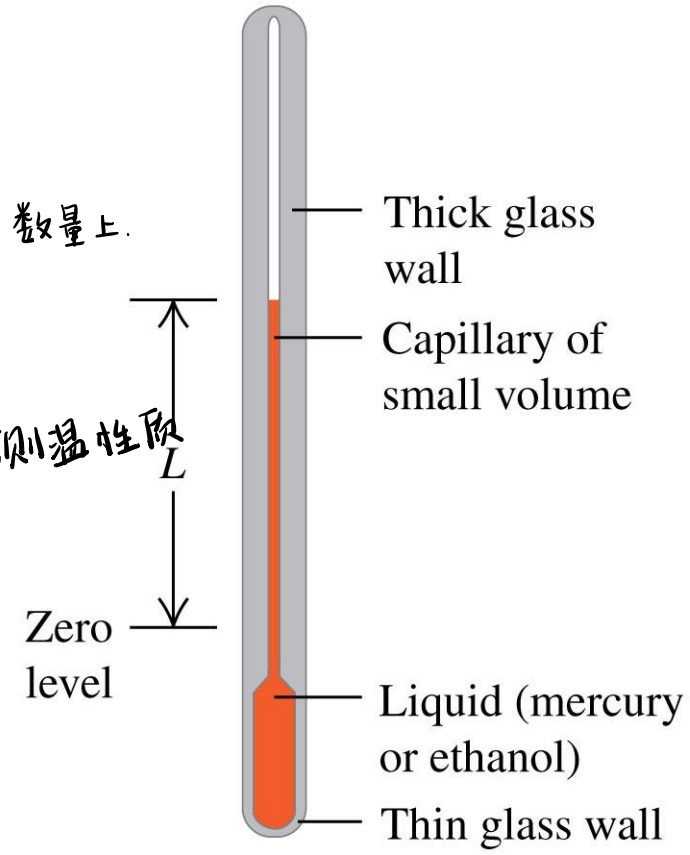
heat
temperature.



How much heat is contained in molten iron at 1500 degrees Celsius?

Temperature

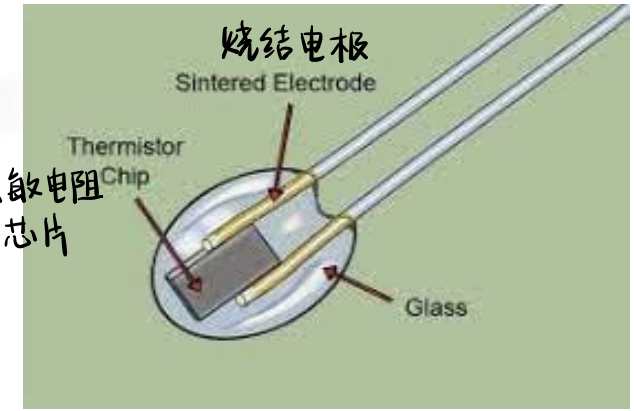
- Can be described as the feeling of hot and cold.
- Temperature depends on the physical state of a material and is a quantitative description of its hotness or coldness.
- Any physical property that varies with temperature is called a thermometric property and can be used to measure temperature, i.e., a thermometer.
- The volume of the thermometric liquid in the thermometer changes with temperature.



Temperature

- Other types of thermometers





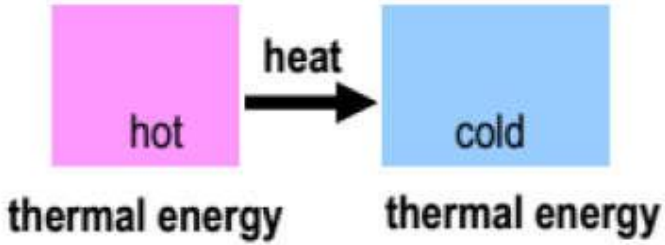
Digital Thermometer



Radiation Thermometer

Thermal Equilibrium 热平衡

- Heat is a thermal energy that flows from a hotter body to a colder body, it is not a static quantity.
- Two systems are in thermal equilibrium if and only if they have the same temperature.



Thermal energy is one of the components of internal energy.
热能属于内能

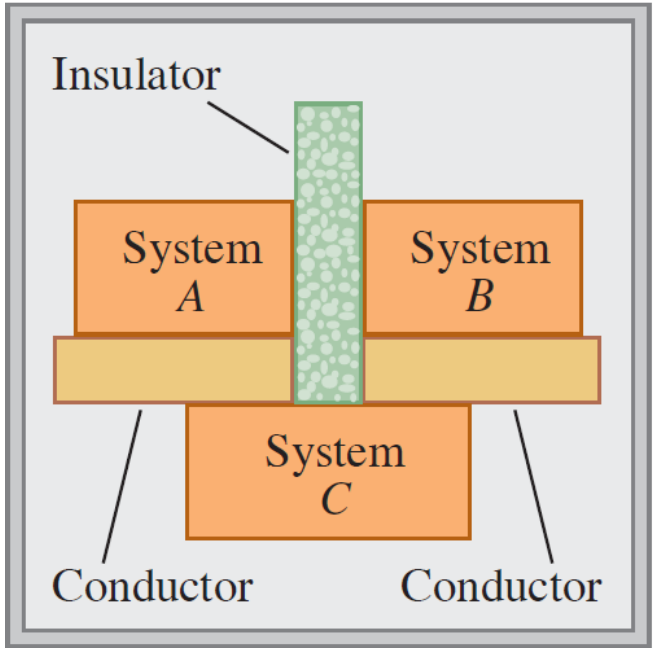
- As long as the two bodies are in thermal contact, heat can flow by the modes of conduction, convection, and radiation.
- When the interaction between two bodies causes no further change in the system, we call this a state of Thermal Equilibrium.

相互作用
热传导，热对流，热辐射

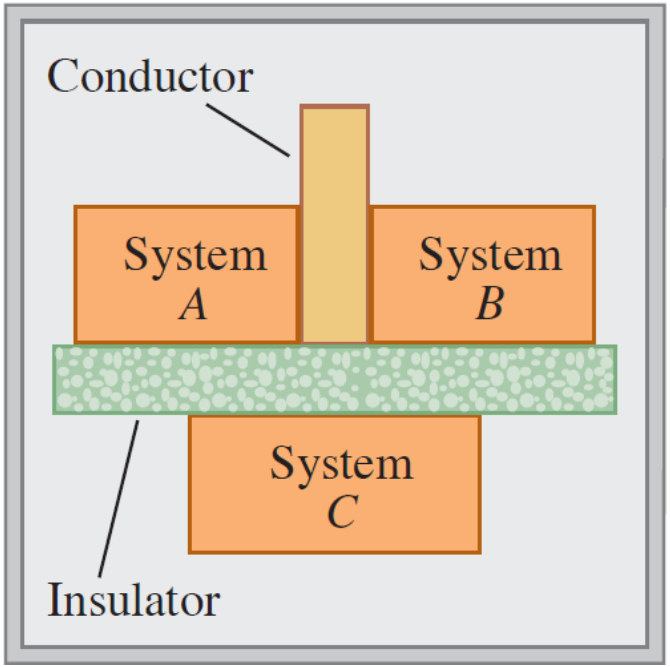
The Zeroth Law of Thermodynamics 热零定律

- If C is initially in thermal equilibrium with both A and B, then A and B are in thermal equilibrium with each other.

(a) If systems *A* and *B* are each in thermal equilibrium with system *C* ...



(b) ... then systems *A* and *B* are in thermal equilibrium with each other.



Temperature Scale

- On the Celsius (or centigrade) temperature scale, 0 degrees Celsius is the freezing point of pure water and 100 degrees Celsius is its boiling point.
- There is a temperature, −273.15 degrees Celsius, at which the absolute pressure of any gas would become zero; aka absolute zero temperature.
- On the Kelvin (or absolute) temperature scale, 0 K is the extrapolated temperature at which a gas would exert no pressure.

$$T_K = T_C + 273.15 \qquad (17.3)$$

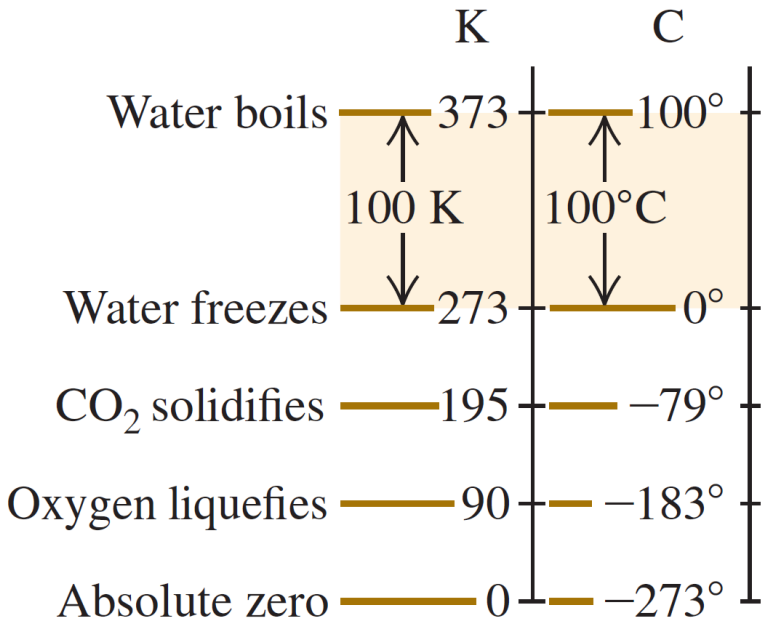
Kelvin temperature — Celsius temperature

Celsius and Kelvin tempreture.
一一对应

Temperature Scale

- Below are relationships between the Kelvin (K) and Celsius (C) temperature scales. Temperatures have been rounded off to the nearest degree.

完成，使完美.



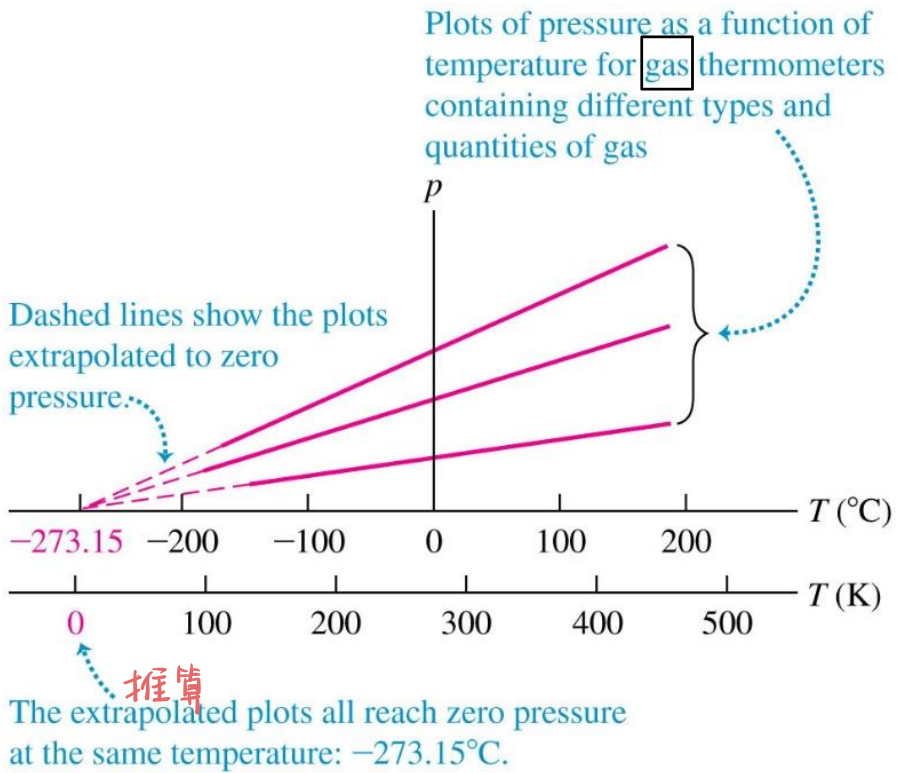
气体温度计

Gas Thermometers and The Kelvin Scale

开尔文刻度

- By extrapolating (推算) the graph, we see that there is a hypothetical (假定) temperature, -273.15°C, at which the absolute pressure of the gas would become zero.

可以发现
不同气体的 P.T. 是线性相关的，
归于一点，仅倾斜角度不同.



Gas Thermometers and The Kelvin Scale

- With a constant volume gas thermometer, the pressure of the constant/fixed volume varies linearly with temperature.
- Slow but accurate, generally used for calibration.

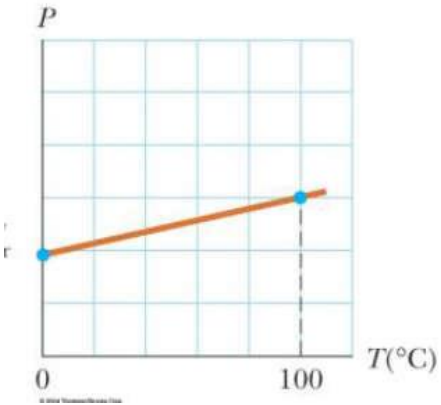
校准

- For Constant Volume Gas Thermometer, the pressure p increase with temperatures, and is directly proportional to the Kelvin temperature.



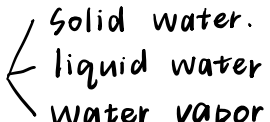
Same gas

$$\frac{T_2}{T_1} = \frac{p_2}{p_1}$$ (constant-volume gas thermometer, T in kelvins) (17.4)



Gas Thermometers and The Kelvin Scale

- To complete the definition of T, we need only specify the Kelvin temperature of a single specific state, i.e., the Triple Point of Water.
- It is the unique combination of temperature and pressure at which solid water (ice), liquid water, and water vapor can all coexist.
- It occurs at a temperature of 0.01 °C and a water vapor pressure of 610 Pa.
- The triple-point temperature of water is defined to have the value corresponding T_{triple} = 273.16 K.
- If p_{triple} is the pressure in a gas thermometer at temperature T_{triple} and p is the pressure at some other temperature T, then T is given on the Kelvin scale by



可以共存的条件 P=610 Pa T=0.01°C =273.16 K. ← 用这个.

$$T = T_{\text{triple}}\frac{p}{p_{\text{triple}}} = (273.16\ \text{K})\frac{p}{p_{\text{triple}}} \qquad (17.5)$$

Thermal Expansion 热胀

- Most materials expand when their temperatures increase.
- For moderate changes in temperature, the change in length is given by:

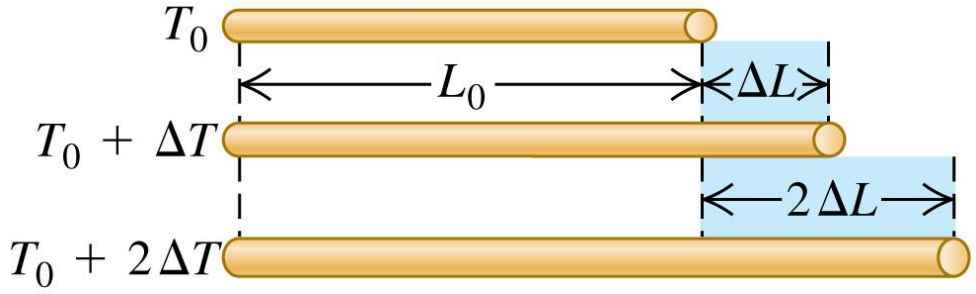
适度

Linear thermal expansion:
Change in length

$$\Delta L = \alpha L_0 \Delta T$$

Original length
Temperature change
Coefficient of linear expansion

(17.6)



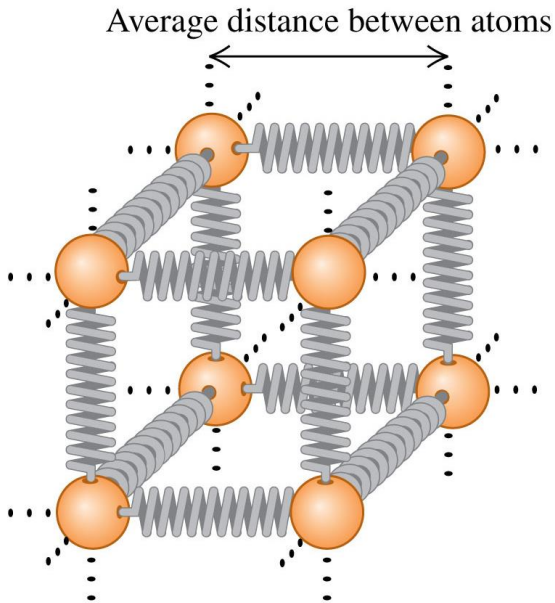
α - coefficient of linear expansion, unit K^{-1}.

- If a body has length L_0 at temperature T_0, then its length L at a temperature $T = T_0 + \Delta T$ is

$$L = L_0 + \Delta L = L_0 + \alpha L_0 \Delta T = L_0(1 + \alpha \Delta T) \quad (17.7)$$

Thermal Expansion

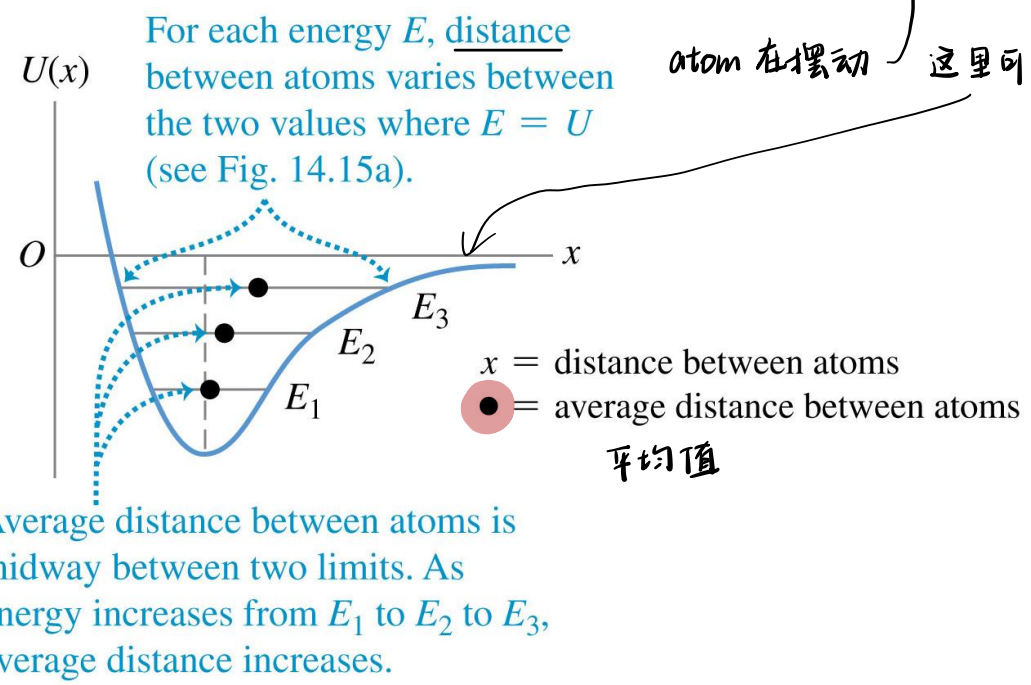
- We can understand linear expansion if we model the atoms as being held together by springs. 类比推理
- When the temperature increases, the average distance between atoms also increases.
- As the atoms get farther apart, every dimension increases.



每个方向的长度都变长

Thermal Expansion

- A graph of the "spring" potential energy versus distance between neighboring atoms is not symmetrical.
- As the energy increases and the atoms oscillate with greater amplitude, the average distance increases.



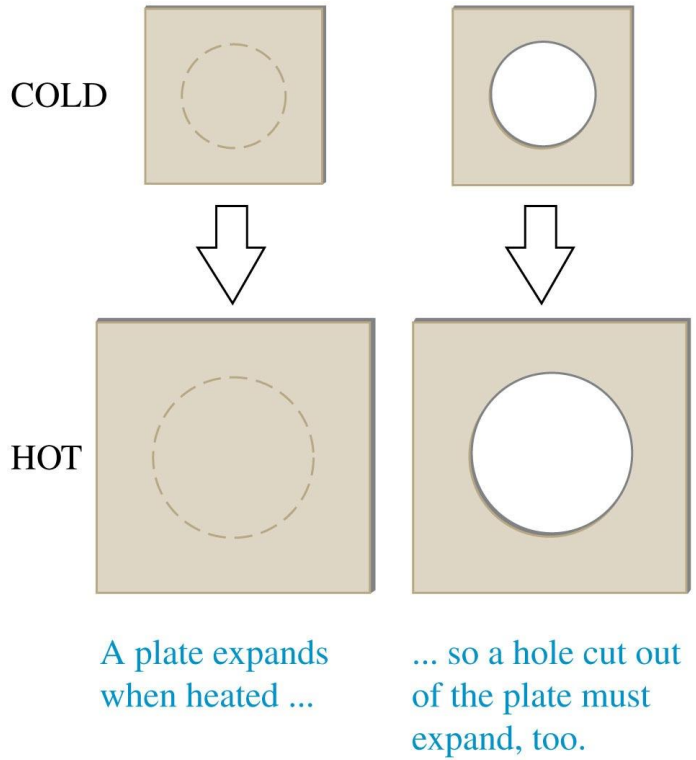
Thermal Expansion

- If an object has a hole in it, the hole also expands with the object, as shown.
- The hole does not shrink.
- The change in volume due to thermal expansion is given by

$$\Delta V = \beta V_0 \Delta T \quad (17.8)$$

where β is the coefficient of volume expansion and is 3 times the coefficient of linear expansion (3α), unit K^{-1}.

$\Delta L = \alpha L_0 \Delta T$. $\beta = 3\alpha$. (K^{-1})



名种材料的α值

Thermal Expansion

TABLE 17.1 Coefficients of Linear Expansion

Material	α [K^{-1} or $(C°)^{-1}$]
Aluminum	2.4×10^{-5}
Brass	2.0×10^{-5}
Copper	1.7×10^{-5}
Glass	$0.4–0.9 \times 10^{-5}$
Invar (nickel–iron alloy)	0.09×10^{-5}
Quartz (fused)	0.04×10^{-5}
Steel	1.2×10^{-5}

各种材料的β值，
对比起来一看，确时是3α.

Thermal Expansion

TABLE 17.2 Coefficients of Volume Expansion

Solids	$\beta\ [\mathrm{K}^{-1}$ or $(\mathrm{C}^\circ)^{-1}]$	Liquids	$\beta\ [\mathrm{K}^{-1}$ or $(\mathrm{C}^\circ)^{-1}]$
Aluminum	7.2×10^{-5}	Ethanol	75×10^{-5}
Brass	6.0×10^{-5}	Carbon disulfide	115×10^{-5}
Copper	5.1×10^{-5}	Glycerin	49×10^{-5}
Glass	$1.2–2.7 \times 10^{-5}$	Mercury	18×10^{-5}
Invar	0.27×10^{-5}		
Quartz (fused)	0.12×10^{-5}		
Steel	3.6×10^{-5}		

Thermal Expansion

EXAMPLE 17.2 LENGTH CHANGE DUE TO TEMPERATURE CHANGE

A surveyor uses a steel measuring tape that is exactly 50.000 m long at a temperature of 20°C. The markings on the tape are calibrated for this temperature. (a) What is the length of the tape when the temperature is 35°C? (b) When it is 35°C, the surveyor uses the tape to measure a distance. The value that she reads off the tape is 35.794 m. What is the actual distance?

校准

(a) $L = \Delta L + L_0 = L_0(1 + \alpha \Delta T)$

$= 50\,[1 + \alpha \times (35°C - 20°C)]$

$= 50 \times (1 + 15 \times 1.2 \times 10^{-5})$

$= 50 + 0.009$

$= 50.009$ m.

(b) $35.794\ m = L_0(1 + \alpha \Delta T)$

$L_0 = \frac{35.794}{1 + 1.2 \times 10^{-5} \times 15} = \frac{35.794}{1.00018}$

EXECUTE: (a) The temperature change is $\Delta T = T - T_0 = 15$ C°; from Eqs. (17.6) and (17.7),

$$\Delta L = \alpha L_0 \Delta T = (1.2 \times 10^{-5}\ \text{K}^{-1})(50\ \text{m})(15\ \text{K})$$

$$= 9.0 \times 10^{-3}\ \text{m} = 9.0\ \text{mm}$$

$$L = L_0 + \Delta L = 50.000\ \text{m} + 0.009\ \text{m} = 50.009\ \text{m}$$

要分清楚计量问题里面

哪个才是原始量

(b) Our result from part (a) shows that at 35°C, the slightly expanded tape reads a distance of 50.000 m when the true distance is 50.009 m. We can rewrite the algebra of part (a) as $L = L_0(1 + \alpha \Delta T)$; at 35°C, *any* true distance will be greater than the reading by the factor $50.009/50.000 = 1 + \alpha \Delta T = 1 + 1.8 \times 10^{-4}$. The true distance is therefore

$$(1 + 1.8 \times 10^{-4})(35.794\ \text{m}) = 35.800\ \text{m}$$

$\Delta V_1 = \beta_{glass} \cdot 200\ cm^3 \cdot \Delta T$

$= 0.4 \times 10^{-5} \times 200 \times 10^{-6}\ m^3 \times (100°C - 20°C)$

$= 64 \times 10^{-8}\ m^3$

$\Delta V_{Hg} = \beta_{Hg} \cdot V_0 \cdot \Delta T$

Thermal Expansion

EXAMPLE 17.3 VOLUME CHANGE DUE TO TEMPERATURE CHANGE

V. 使满溢　　水银

A 200-cm^3 glass flask is filled to the brim with mercury at 20°C. How much mercury overflows when the temperature of the system is raised to 100°C? The coefficient of *linear* expansion of the glass is $0.40 \times 10^{-5}\ K^{-1}$.

给的是 α，不是 β

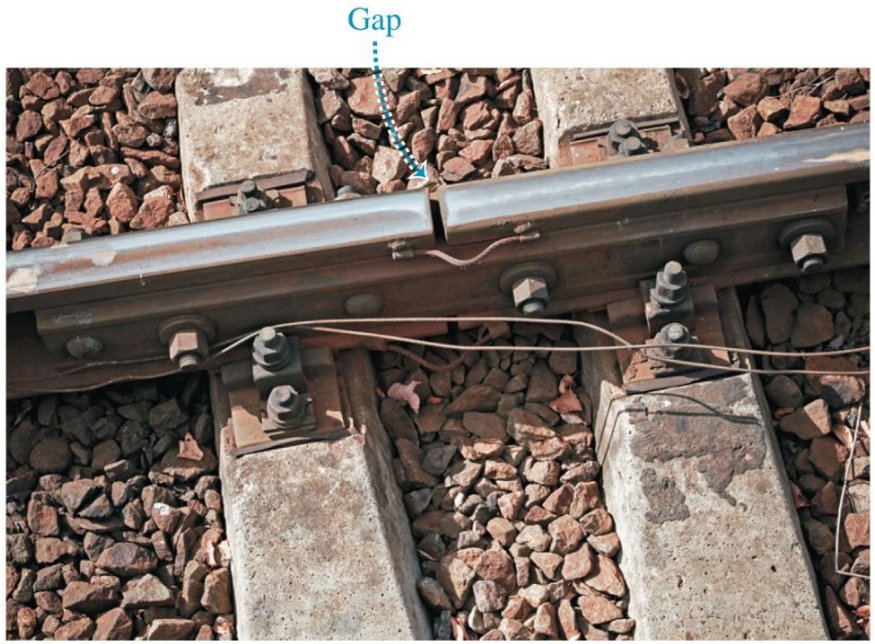
EXECUTE: From Table 17.2, $\beta_{Hg} = 18 \times 10^{-5}\ K^{-1}$. That is indeed greater than $\beta_{glass} = 3\alpha_{glass} = 3(0.40 \times 10^{-5}\ K^{-1}) = 1.2 \times 10^{-5}\ K^{-1}$, from Eq. (17.9). The volume overflow is then

$$\begin{aligned}\Delta V_{Hg} - \Delta V_{glass} &= \beta_{Hg} V_0 \Delta T - \beta_{glass} V_0 \Delta T \\ &= V_0 \Delta T(\beta_{Hg} - \beta_{glass}) \\ &= (200\ cm^3)(80\ C°)(18 \times 10^{-5} - 1.2 \times 10^{-5}) \\ &= 2.7\ cm^3\end{aligned}$$

This is the working principle of the mercury-in-glass thermometer.

Thermal Expansion

- This railroad track has a gap between segments to allow for thermal expansion.
- On hot days, the segments expand and fill in the gap.
- If there were no gaps, the track could buckle under very hot conditions.



Thermal Expansion – Thermal Stress

- If we change the temperature of a rod 棒、条 but prevent it from expanding or contracting, thermal stress develops.

这是计算的条、棒的 thermal stress.

Thermal stress:

Force needed to keep length of rod constant

$$\frac{F}{A} = -Y\alpha\,\Delta T \qquad (17.12)$$

- F: Force needed to keep length of rod constant
- A: Cross-sectional area of rod 截面积
- Y: Young's modulus 系数
- α: Coefficient of linear expansion
- ΔT: Temperature change

- Expansion joints 关节 on bridges are needed to accommodate 容纳 changes in length that result from thermal expansion.

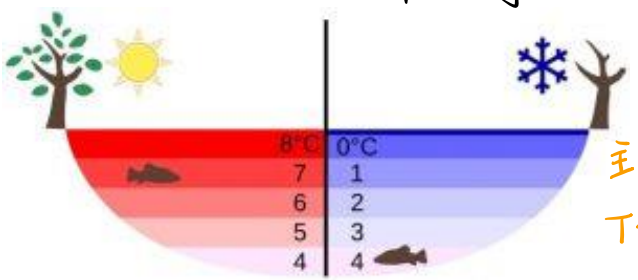


比热容（Specific Heat Capacity），用符号c表示，又称比热容量，简称比热（specific heat），是单位质量物质的热容量，即单位质量物体改变单位温度时吸收或放出的热量。

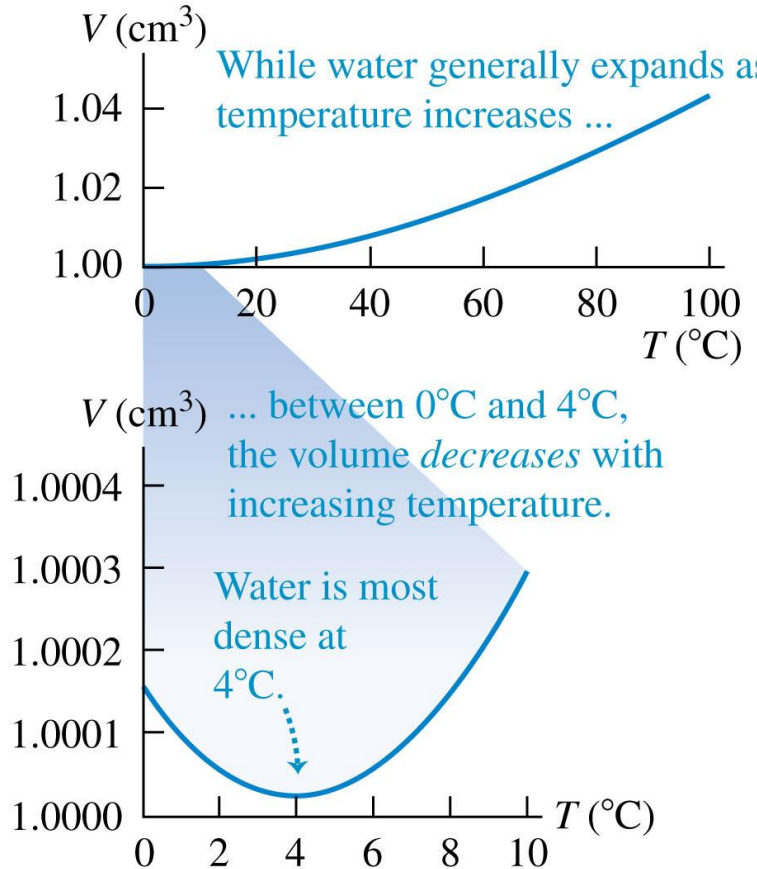
Thermal Expansion

- Thermal Expansion of Water
- Between 0 degrees Celsius and 4 degrees Celsius, water decreases in volume with increasing temperature.
- Because of this anomalous behavior, lakes freeze from the top down instead of from the bottom up.

density ↑ ρ↑ V↓ ρ↑

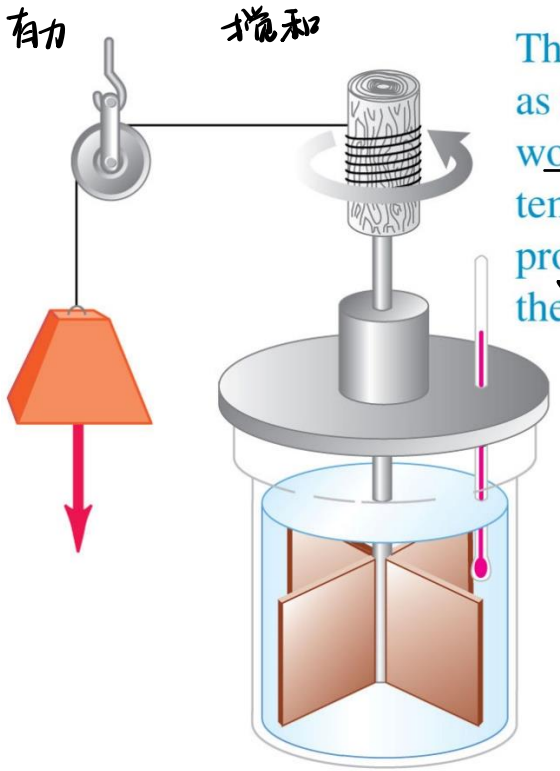


主要是密度.在0~4℃
T↑ V↓ ρ↑ 温度高的水沉在底部



Quantity of Heat

- Sir James Joule (1818–1889) studied how water can be warmed by vigorous stirring with a paddle wheel.

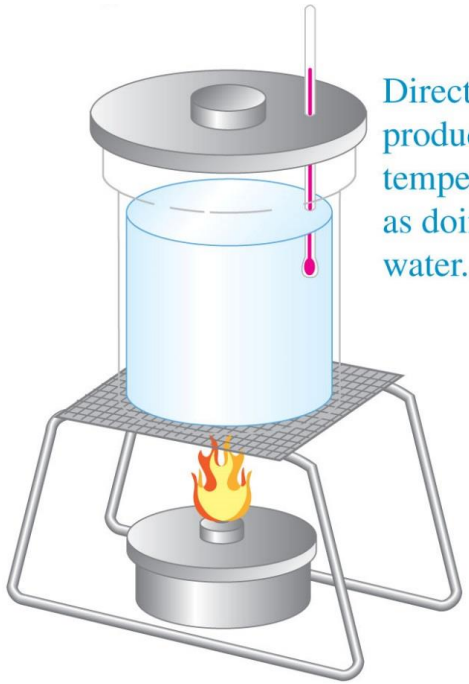


The water warms as the paddle does work on it; the temperature rise is proportional to the amount of work done.

水温的升高与做功是呈比例关系的.

Quantity of Heat 热量

- Energy transfer that takes place solely 唯一的 because of a temperature difference is called heat flow or heat transfer, and energy transferred in this way is called heat. is a form of energy.
- The same temperature change caused by stirring can also be caused by putting the water in contact with some hotter object.
- The calorie (abbreviated cal) is the amount of heat required to raise the temperature of 1 gram 克 of water from 14.5 degrees Celsius to 15.5 degrees Celsius.



Direct heating can produce the same temperature change as doing work on the water.

两种

Quantity of Heat

华氏温标

- A corresponding unit of heat using Fahrenheit degrees and British units is the British thermal unit, or Btu.
- One Btu is the quantity of heat required to raise the temperature of 1 pound (weight) of water 1°F from 63°F to 64°F.
- Calorie or Btu are not the fundamental SI unit for heat.
- The SI unit for heat (or other types of energy) is Joule.

之间的能量转换

$$1\ \text{cal} = 4.186\ \text{J}$$

$$1\ \text{kcal} = 1000\ \text{cal} = 4186\ \text{J}$$

$$1\ \text{Btu} = 778\ \text{ft} \cdot \text{lb} = 252\ \text{cal} = 1055\ \text{J}$$

air con…

热容的标准定义是：“当一系统由于加给一微小的热量dQ而温度升高dT时，dQ/dT 这个量即是该系统的热容。”（GB3102.4-93），通常以符号C表示，单位J/K

Heat Capacity 热容，C

- The heat capacity (C) of a body is the amount of heat required to raise the temperature of that body by one degree of temperature.
- If the heat capacity varies with temperature, we can define heat capacity as the ratio of the infinitesimal heat flow (dQ) to the infinitesimal temperature change (dT):

$\left(C = \frac{dQ}{dT} \right)$ unit of heat capacity is J /ºC, which is identical to J K^{-1} .

Heat capacity is an extensive property, i.e., it depends on size of the body.

$C = \frac{dQ}{dT}$ J/K

比热容（Specific Heat Capacity），用符号c表示，又称比热容量，简称比热（specific heat），是单位质量物质的热容量，即单位质量物体改变单位温度时吸收或放出的热量。

Specific Heat

- The quantity of heat Q required to increase the temperature of a mass m of a certain material by ΔT is:

Heat required to change temperature of a certain mass $Q = mc\Delta T$ (17.13)

Mass of material; Temperature change; Specific heat of material

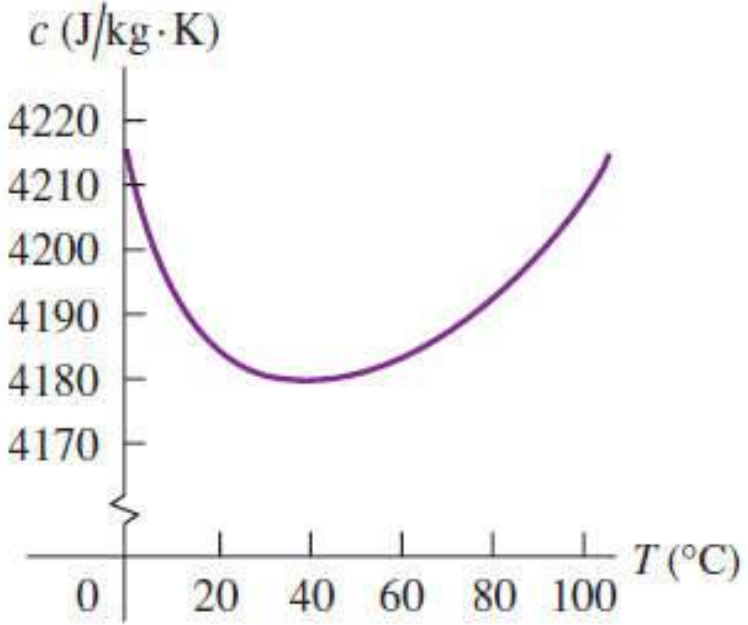
- The specific heat or specific heat capacity *c* of a substance is its heat capacity per unit mass, it does not depend on the size of the body.
- Different materials has different specific heat values.
- The specific heat of water is approximately 4190 J/kg·K.

水的比热容不是定值，我们取值 4190 J/kg·K

Specific Heat of water

17.17 Specific heat of water as a function of temperature. The value of c varies by less than 1% between 0°C and 100°C.

不是直线.



Specific Heat of water



17.18 Water has a much higher specific heat than the glass or metals used to make cookware. This helps explain why it takes several minutes to boil water on a stove, even though the pot or kettle reaches a high temperature very quickly.

摩尔比热容.

Molar Heat Capacity

- The quantity of heat Q required to increase the temperature of n moles of a certain material by ΔT is:

num of moles →

Heat required to change temperature of a certain number of moles ⋯→ $Q = nC\,\Delta T$ (17.18)

n: Number of moles of material
ΔT: Temperature change
C: Molar heat capacity of material

Molar heat capacity

- The molar mass of any substance, denoted by *M*, is the mass per mole.

摩尔质量 物质

$$m = nM \qquad Q = nMc\,\Delta T \longleftarrow C = Mc$$

$Q = nC\,\Delta T$

- The molar heat capacity *C* has different values for different materials. The molar heat capacity of water is approximately 75.4 J/mol·K.

Molar Heat Capacity

TABLE 17.3 Approximate Specific Heats and Molar Heat Capacities (Constant Pressure)

Substance	Specific Heat, c (J/kg · K)		Molar Mass, M (kg/mol)		Molar Heat Capacity, C (J/mol · K)
Aluminum	910	×	0.0270	=	24.6
Beryllium	1970		0.00901		17.7
Copper	390		0.0635		24.8
Ethanol	2428		0.0461		111.9
Ethylene glycol	2386		0.0620		148.0
Ice (near 0°C)	2100		0.0180		37.8
Iron	470		0.0559		26.3
Lead	130		0.207		26.9
Marble ($CaCO_3$)	879		0.100		87.9
Mercury	138		0.201		27.7
Salt (NaCl)	879		0.0585		51.4
Silver	234		0.108		25.3
Water (liquid)	4190		0.0180		75.4

Other Specific Heat Capacities

- Measurements for solid materials are usually made at constant atmospheric pressure; the corresponding values are called the specific heat and molar heat capacity at constant pressure, denoted by cp and Cp.
- For a gas it is usually easier to keep the substance in a container with constant volume; the corresponding values are called the specific heat and molar heat capacity at constant volume, denoted by cV and CV.
- For a given substance, CV and Cp are different. (If the system can expand while heat is added, there is additional energy exchange through the performance of work by the system on its surroundings. If the volume is constant, the system does no work.)

Phase Changes

- The phases (or states) of matter are solid, liquid, and gas.
- A phase change is a transition from one phase to another.
- The temperature does not change during a phase change.
- The latent heat, L, is the heat per unit mass that is transferred in a phase change.

Heat transfer in a phase change: $Q = \pm mL$ (17.20)

- m: Mass of material that changes phase
- L: Latent heat for this phase change
- $+$ if heat enters material, $-$ if heat leaves

ice melt

water → ice

改变物质状态所需吸热量

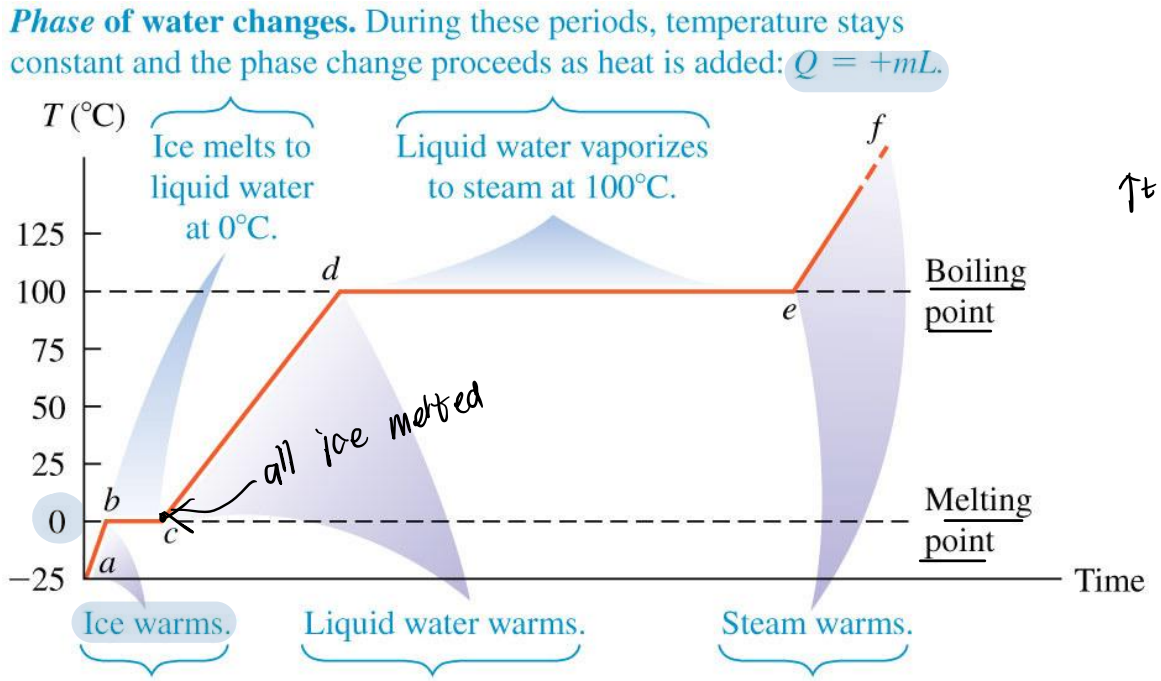


filled with ice initially 0°C

Phase Changes

tempreture keep when phase changes

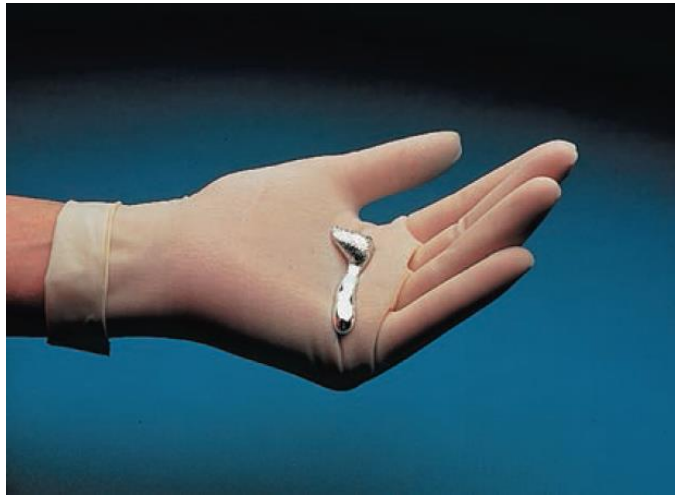
- When heat is added to ice at a constant rate



↑t ↑Q

Phase Changes

- The heat of fusion is the amount of heat required to melt the frozen solvent.
- The metal gallium, shown here melting in a person's hand, is one of the few elements that melts at room temperature.
- Its melting temperature is 29.8 degrees Celsius, and its heat of fusion is $L_f = 8.04 \times 10^4$ J/kg.



Phase Changes

- The heat of vaporization is the amount of heat needed to turn liquid into a vapor.
- The water may be warm and it may be a hot day, but these children will feel cold when they first step out of the swimming pool.
- That's because as water evaporates from their skin, it removes the heat of vaporization from their bodies.
- To stay warm, they will need to dry off immediately.



Mechanisms of Heat Transfer 传热机理

- In nature, energy naturally flows from higher temperature objects to lower temperature objects; this is called heat transfer.
- The three mechanisms of heat transfer are conduction, convection, and radiation.
- Conduction occurs within an object or between two objects in contact.
- Convection 对流 depends on motion of mass from one region of space to another.
- Radiation is heat transfer by electromagnetic radiation 电磁, such as sunshine, with no need for matter to be present in the space between objects.

Mechanisms of Heat Transfer – Conduction

- In conduction, heat flows from a higher to a lower temperature.
- Consider a solid rod of conducting material with cross-sectional area A and length L.
- The left end of the rod is kept at a temperature T_H and the right end at a lower temperature T_C.
- The rate that heat is transferred is:

熱流

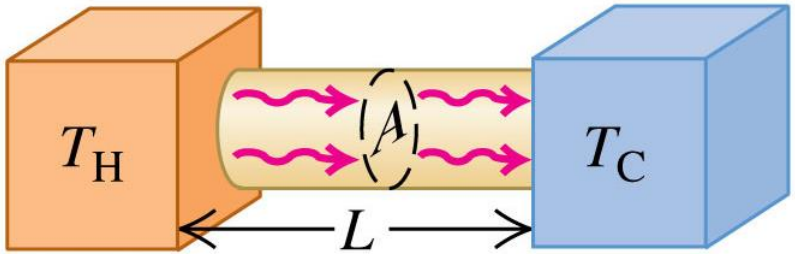
Rate of heat flow — Temperatures of hot and cold ends of rod

Heat current in conduction

$$H = \frac{dQ}{dt} = \left(kA \frac{T_H - T_C}{L} \right)$$

Length of rod (17.21)

Thermal conductivity of rod material — Cross-sectional area of rod



k↑ Q↑

Thermal Conductivities of Some Common Substances

TABLE 17.5 Thermal Conductivities

Substance	k (W/m · K)
Metals	
Aluminum	205.0
Brass	109.0
Copper	385.0
Lead	34.7
Mercury	8.3
Silver	406.0
Steel	50.2

TABLE 17.5 Thermal Conductivities

Substance	k (W/m · K)
Solids (representative values)	
Brick, insulating	0.15
Brick, red	0.6
Concrete	0.8
Cork	0.04
Felt	0.04
Fiberglass	0.04
Glass	0.8
Ice	1.6
Rock wool	0.04
Styrofoam	0.027
Wood	0.12–0.04

TABLE 17.5 Thermal Conductivities

Substance	k (W/m · K)
Gases	
Air	0.024
Argon	0.016
Helium	0.14
Hydrogen	0.14
Oxygen	0.023

Mechanisms of Heat Transfer – Conduction

- If the temperature varies in a nonuniform way along the length of the conducting rod, we introduce a coordinate x along the length and generalize the temperature gradient to be dT/dx:

$$H = \frac{dQ}{dt} = -kA\frac{dT}{dx} \qquad (17.22)$$

- The negative sign indicates that heat flows in the direction of decreasing temperature.
- If temperature increases with increasing x, then dT/dx > 0 and H < 0; the negative value of H in this case means that heat flows in the negative x-direction, from high to low temperature.

热阻指的是当有热量在物体上传输时，在物体两端温度差与热源的功率之间的比值。单位为开尔文每瓦特（K/W）或摄氏度每瓦特（°C/W）。当热量流过两个相接触的固体的交界面时，界面本身对热流呈现出明显的热阻，称为接触热阻

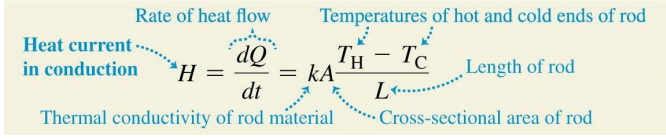
Mechanisms of Heat Transfer – Conduction

- For thermal insulation in buildings, engineers use the concept of thermal resistance, denoted by R.
- The thermal resistance R of a slab (厚板) of material with area A is defined so that the heat current H through the slab is

$$H = \frac{A(T_H - T_C)}{R} \quad (17.23)$$

- where T_H and T_C are the temperatures on the two sides of the slab, and

$$R = \frac{L}{k} \quad (17.24)$$



- where L is the thickness of the slab. The SI unit of R is 1 m^2.K/W.

Mechanisms of Heat Transfer – Conduction

EXAMPLE 17.11 CONDUCTION INTO A PICNIC COOLER

A Styrofoam cooler (**Fig. 17.25a**) has total wall area (including the lid) of 0.80 m^2 and wall thickness 2.0 cm. It is filled with ice, water, and cans of Omni-Cola, all at 0°C. What is the rate of heat flow into the cooler if the temperature of the outside wall is 30°C? How much ice melts in 3 hours?

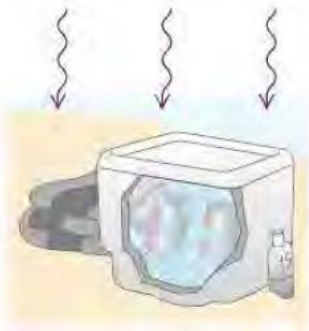
$H = kA\frac{T_H - T_C}{L} = 0.027\ \text{W/m·K} \times 0.8\ m^2 \times \frac{30°C - 0°C}{0.02\ m}$

$= 32.4\ \text{J/s}$ $\quad Q = 32.4\ \text{J/s} \times 3 \times 60 \times 60 = \text{mL}$

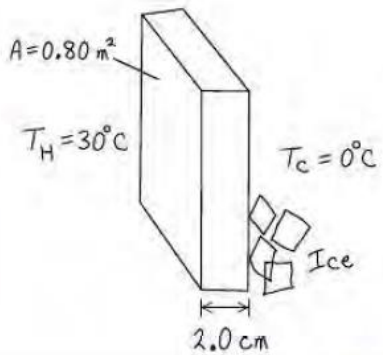
$L_f = 3.34 \times 10^5\ \text{J/kg}$

17.25 Conduction of heat across the walls of a Styrofoam cooler.

(a) A cooler at the beach



(b) Our sketch for this problem



EXECUTE: We assume that the total heat flow is the same as it would be through a flat Styrofoam slab of area 0.80 m^2 and thickness 2.0 cm = 0.020 m (Fig. 17.25b). We find k from Table 17.5. From Eq. (17.21),

$$H = kA\frac{T_H - T_C}{L} = (0.027\ \text{W/m·K})(0.80\ \text{m}^2)\frac{30°\text{C} - 0°\text{C}}{0.020\ \text{m}}$$

$$= 32.4\ \text{W} = 32.4\ \text{J/s}$$

The total heat flow is $Q = Ht$, with $t = 3\ \text{h} = 10{,}800\ \text{s}$. From Table 17.4, the heat of fusion of ice is $L_f = 3.34 \times 10^5\ \text{J/kg}$, so from Eq. (17.20) the mass of ice that melts is

$$m = \frac{Q}{L_f} = \frac{(32.4\ \text{J/s})(10{,}800\ \text{s})}{3.34 \times 10^5\ \text{J/kg}} = 1.0\ \text{kg}$$

Mechanisms of Heat Transfer – Convection

- Convection is the transfer of heat by the mass motion of fluid. 流动的
- For example: hot-air and hot-water home heating systems, the cooling system of an automobile engine, and the flow of blood in the body.
- A heating 加热元件 element in the tip of this submerged 浸没 tube warms the surrounding water, producing a complex pattern of free convection.



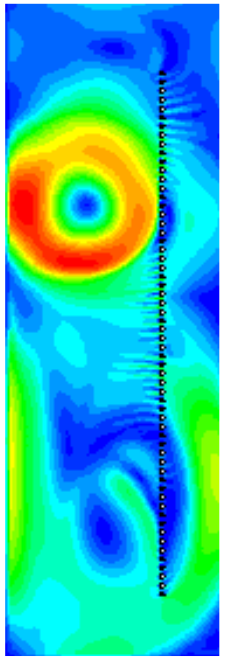
Mechanisms of Heat Transfer – Convection

自然对流

- Natural convection is when the flow of fluid occurs because of the difference in density caused by thermal expansion such as hot air rising.

强对流

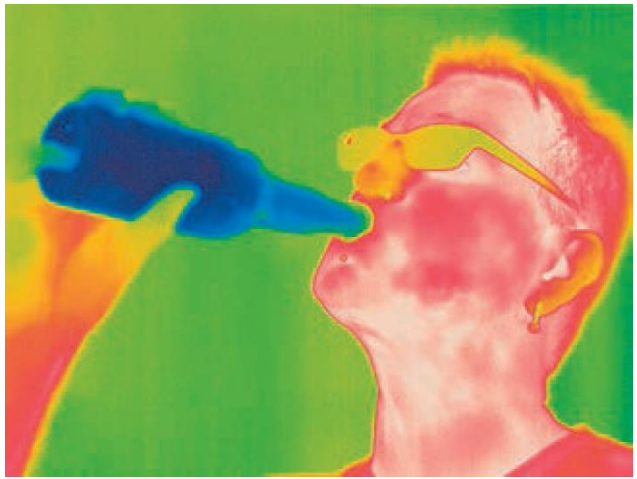
- Force convection is when the flow of the fluid occurs because it is forced by a blower or pump, i.e., the most important mechanism for heat transfer within the human body (needed to maintain nearly constant temperature in various environments) is forced convection of blood, with the heart serving as the pump.
- Convective heat transfer is a very complex process, no simple equation to calculate.



Using Fluent Software to simulate heat pipe for heat dissipation

Mechanisms of Heat Transfer – Radiation

- Radiation is the transfer of heat by electromagnetic waves, such as visible light or infrared.
- This false-color infrared photograph reveals radiation emitted by various parts of the man's body. The strongest emission comes from the warmest areas, while there is very little emission from the bottle of cold beverage.



Mechanisms of Heat Transfer – Radiation

- Stefan-Boltzmann law state that the rate at which a surface radiates electromagnetic energy is proportional to the fourth power of its absolute temperature.
- The heat current in radiation is:

$$\sigma = 5.670373(21) \times 10^{-8}\ \mathrm{W/m^2 \cdot K^4}$$

Stefan–Boltzmann constant

Heat current in radiation

$$H = Ae\sigma T^4$$

Area of emitting surface

Emissivity of surface

Absolute temperature of surface

- Emissivity, e is a dimensionless number between 0 and 1, it represents the ratio of the rate of radiation from a particular surface to the rate of radiation from an equal area of an ideal radiating surface at the same temperature.
- If e = 1, we say the body is a blackbody. Otherwise referred as graybody.

Mechanisms of Heat Transfer – Radiation

EXAMPLE 17.14 HEAT TRANSFER BY RADIATION

A thin, square steel plate, 10 cm on a side, is heated in a blacksmith's forge to 800°C. If the emissivity is 0.60, what is the total rate of radiation of energy from the plate?

$H = Ae\sigma \cdot T^4$
$= 2(0.1)^2 \times 0.6 \times 5.67 \times 10^{-8} \times (1073)^4$

EXECUTE: The total surface area is $2(0.10\ \text{m})^2 = 0.020\ \text{m}^2$, and $T = 800°\text{C} = 1073\ \text{K}$. Then Eq. (17.25) gives

求物体幅射率

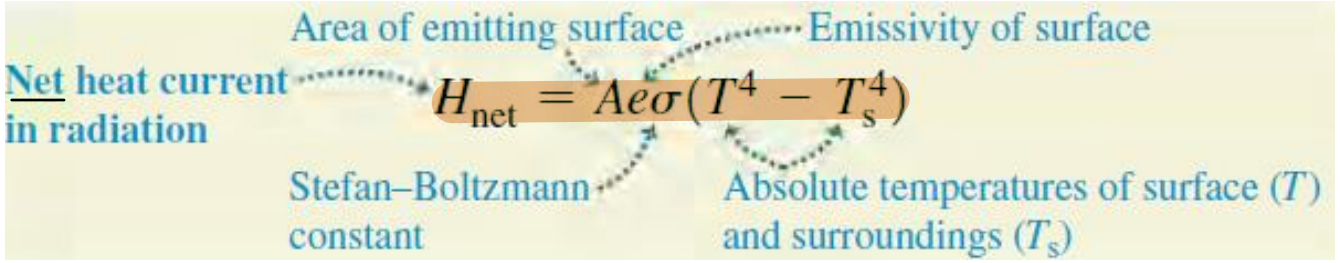
$$H = Ae\sigma T^4$$
$$= (0.020\ \text{m}^2)(0.60)(5.67 \times 10^{-8}\ \text{W/m}^2 \cdot \text{K}^4)(1073\ \text{K})^4$$
$$= 900\ \text{W} = 900\ \text{J/s}$$

EVALUATE: The nearby blacksmith will easily feel the heat radiated from this plate.

Mechanisms of Heat Transfer – Radiation

Radiation and Absorption

- While a body at absolute temperature T is radiating, its surroundings at temperature Ts are also radiating, and the body absorbs some of this radiation.
- If it is in thermal equilibrium with its surroundings, T = Ts and the rates of radiation and absorption must be equal, then the rate of absorption must be $H = Ae\sigma T_s^4$
- The net rate of radiation from a body at temperature T with surroundings at temperature Ts



Mechanisms of Heat Transfer – Radiation

EXAMPLE 17.15 RADIATION FROM THE HUMAN BODY

What is the total rate of radiation of energy from a human body with surface area 1.20 m^2 and surface temperature 30°C = 303 K? If the surroundings are at a temperature of 20°C, what is the *net* rate of radiative heat loss from the body? The emissivity of the human body is very close to unity, irrespective of skin pigmentation.

273 + 20 = 293 K

$H = Ae\sigma T^4$
$= 1.2 \times 1 \times 5.67 \times 10^{-8} \times 303^4$
$= 574$ W

$H_{net} = Ae\sigma(T^4 - T_s^4) = 72$ W

EXECUTE: Taking $e = 1$ in Eq. (17.25), we find that the body radiates at a rate

$H_s = Ae\sigma T_s^4$

$$H = Ae\sigma T^4$$
$$= (1.20\ \text{m}^2)(1)(5.67 \times 10^{-8}\ \text{W/m}^2 \cdot \text{K}^4)(303\ \text{K})^4 = 574\ \text{W}$$

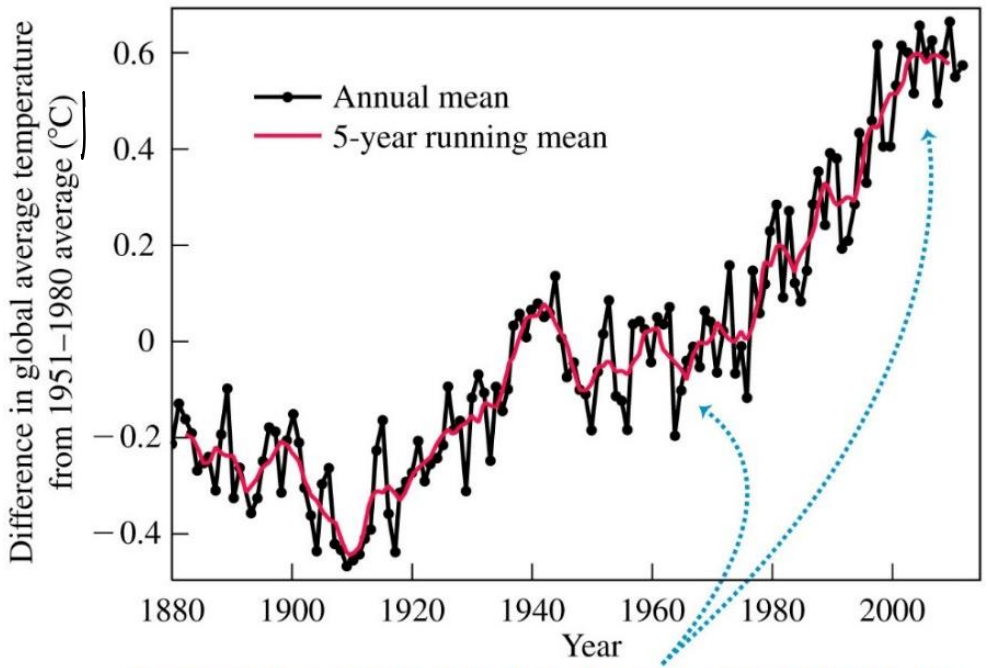
This loss is partly offset by absorption of radiation, which depends on the temperature of the surroundings. From Eq. (17.26), the *net* rate of radiative energy transfer is

$$H_{\text{net}} = Ae\sigma(T^4 - T_s^4)$$
$$= (1.20\ \text{m}^2)(1)(5.67 \times 10^{-8}\ \text{W/m}^2 \cdot \text{K}^4)$$
$$\times [(303\ \text{K})^4 - (293\ \text{K})^4]$$
$$= 72\ \text{W}$$

EVALUATE: The value of H_{net} is positive because the body is losing heat to its colder surroundings.

Radiation and Climate Change

- The energy radiated by the earth's surface is mostly infrared. 红外线的
- CO_2 molecules 分子 in our atmosphere readily absorb some of this infrared radiation and reradiate part of it back down toward the surface.



Increased atmospheric CO_2 due to burning of fossil fuels is the cause of this continuing increase in global average temperatures.

Summary

- Temperature and Heat has been studied.
  - Basic concepts and principles are discussed with detailed analysis and examples
    1. **Temperature and Heat**
    2. **Thermometers and Temperature Scale**
    3. **Gas Thermometers and Kelvin Scale**
    4. **Thermal Expansion**
    5. **Quantity of Heat**
    6. **Phase changes**
    7. **Mechanisms of Heat Transfer**
- Textbook
  - Chapter 17：Temperature and Heat

ΔT

°C / K

T^4

need to use K.